

Linux is its Object Storage Architecture which can split a particular file from its metadata and save them individually. It also provides support for Object Storage Device (OSD) specification along with the Linux clUSTER (Lustre) and Extended Object File System (exofs).

## Finding out all storage devices connected

As discussed earlier, Linux provides a robust storage platform. It supports nearly all types of storage devices in dedicated as well as network mode. Depending upon the type of storage device, there are different commands to track them. Linux supersedes the capabilities, in terms of storage devices, relative to other operating systems. Be it physical storage such as hard disks or network storage or virtual storage, Linux has the potential to deal with all types.

Details of storage devices are stored in the `/dev` directory in Linux. To know the storage devices connected to your system, you must have access to this directory. Once you have the access, following command lists all types of storage devices attached with your system -

```
[me@linuxbox ~]$ ls /dev
```

This command lists all storage devices. For identification of the type of storage device, following is the key -

- The floppy drive is named as `/dev/fd`.
- The SCSI disk (SCSI ID address-wise) is named as `/dev/sda`.
- The SCSI CD-ROM is named as `/dev/scd` or as `/dev/sr`.
- The master disk on the IDE primary controller is named `/dev/hda`.
- The slave disk on the IDE primary controller is named `/dev/hdb`.
- The master and slave disks of the secondary controller are marked as `/dev/hdc` and `/dev/hdd`, respectively.

Device names are automatically appended by 1 in case there are multiple devices of the same category. Symbolic links like `/dev/cdrom` or `/dev/dvd` or `/dev/floppy` are also adopted for convenience to point out the particular devices.

In case your system does not mount removable devices automatically, the following command can help you detect the devices when it is attached-

```
[me@linuxbox ~]$ sudo tail -f /var/log/messages
```

The last few lines of the file will be displayed, and then it pauses. Then, you may insert your removable storage device and check its connectivity.

## Creating new file systems

Any device, be it a dedicated storage device connected to your terminal or any network associated storage device, Linux works on a directory-tree format.